

Professional Summary

Data Scientist professional with 12+ years of total experience in developing and **implementing data-driven solutions** for business problems. With 6+ years of proven leadership. Proficient in **statistical analysis, machine learning algorithms, and data visualization tools**. Skilled in programming languages such as **Python and R** and experienced in working with **data analysis tools**.

Skills & Abilities

Technical Skills: Statistical Analysis, **Machine Learning** Algorithms, Data Visualization, Python, R programming, SQL, **Big data technologies**, Data cleaning and preprocessing, **Deep learning** algorithms, **Natural Language Processing**, RLDA, F&DT, Accelerated Testing

Tools and Software: Azure DevOPS, Github, Azure Data Factory, Azure Foundry, Azure Machine Learning, Python, Mlflow

Experience

Manager Projects
Cognizant

Feb 2024- Present
Bangalore, India

- Leading the team of 8 members, skilled in **Azure Machine Learning, Azure Foundry**, Azure Data factory, Databricks, Synapse SQL, **Pyspark**, Blob Storage and ADLS Gen2.
- Built an end-to-end annotation and synchronization tool integrating video frames, spectral logs, and key procedure event data.
- Implemented video-spectral mapping enabling precise alignment of 2048 spectral wavelengths with video frames • Developed Gaussian smoothing and spectral peak detection logic in the 800- 830nm range with dynamic threshold (>30% of global max intensity).
- Created EDA dashboards with interactive visualizations: time series plots, box plots, cluster heatmaps, and histogram analysis.
- Designed and deployed a model monitoring dashboard using Fabric with key ML metrics: ROC Curve, Confusion Matrix, Train vs Test Error, Sensitivity and Confidence Score.
- Visualized peak regions with highlighted spectra using matplotlib and interpreted key signal events(white light/dark light)
- Used OpenCV for video processing (frame extraction, resizing, RGBA conversion) to support browser based interactive analysis.
- Cleaned and processed large time series CSV datasets for ML model input validation workflows.
- Developed a real-time, RAG-based system that uses a Vector Database and Video Language Model (VLM) to provide surgeons with context-aware insights by analyzing surgical video and voice annotations.
- Created a React + Electron Windows application that integrates a KNN machine learning algorithm to provide real- time laser data overlay, boosting surgical efficiency by 10-30%.
- Built a multimodal system using LangChain and a Vector Database that recommends dome shapes for hearing aids, improving clinical decision-making reliability.

Manager Deputy
Forvia

Jan 2013 – Feb 2024
Bangalore, India

- Analysed large datasets using statistical methods to identify trends, patterns and provided insights to inform business decisions.
- Collaborated with cross-functional teams to understand business requirements and translate them into data-driven solutions.

- To conduct KPI evaluation, performance evaluation, implementing and optimizing the work flow processes and procedures, workload forecast, team utilization, career mapping, interviewing new talents.
 - Mentoring of new hires and assigning career trajectory, supervising and coaching technical report writing, implement test optimization strategies to improve efficiency and reduce lead time.
 - Sensor Database Development, resulting in 30% time reduction in RFQ stage design development cycle, resulting in savings of 2million USD yearly.
 - Road Load Acquisition of time history data on transit van, SUV and crossover; Strain Gage Calibration of After Treatment System, working on component validation using test samples, component rig test and engine test data through testing equipment's.
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- Development of 'Test Failure Predictor' tool to forecast future trends to mitigate test failures, resulting in 1million USD savings yearly.
 - Monitoring and controlling the hydraulic oscillating head, develop or improve processes for physical testing and simulation, structural Fatigue testing and SN Curve development.
 - Provided guidance and mentorship to team members to improve their technical skills and productivity, participated in the development of test plans and protocols.
 - Supporting root cause analysis efforts to identify and address failure modes, conducting reliability testing and analysing test data to assess product performance.
 - Responsible for conducting analysis of road load data time history data, fatigue life estimation, block load and PSD development, engine resonance analysis, and full system durability. **Projects**
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AI-Powered Detection of Kidney Stone Composition: Determination of kidney stone intraoperatively in real-time based on spectral data using **KNN classifier algorithm achieved 72.6% sensitivity**.

Tech Stack: AzureML, ADF, Python, Power BI, Fabric, Machine Learning, Data Preprocessing.

Care Plan Proposer: By leveraging the patient demographics, HRA, medical records and claim history, the **RAG system** generates a comprehensive nutrition, physical exercise and medication plan aiding the health risk prevention. Tech Stack: Python, FAISS, Azure Open AI Service, LangChain, React, Node.js

Part Failure Mitigator: Developed a **decision tree model** for identifying critical features influencing part failure resulting in forecasting of future trends such as engine thermal shock test failure, by **six sigma** approach. Tech Stack: Python, Excel, Machine Learning, Data Preprocessing.

Price Evolution: Improved overall decision-making of vendor product selection by **25%** by designing and developing **BI reports** and providing invaluable insights on price evolution by using **visualizing and analytical tools**. Tech Stack: Power BI, Python, Data Preprocessing

Load Predictor: Methodology to predict the load of any component based on the historical physical testing data. Using the **random forest regression model** got the **96.8% accuracy** and performance on the data. Tech Stack: Excel, Python, Machine Learning, Data Mining

Temperature Mapping: To predict the temperature profile on a muffler/ catalytic converter based on the historical RLDA data. Using the **MLP regressor** artificial neural network model got the **R2 score of 93.7%**. Tech Stack: Excel, Python, Machine Learning, mySQL, Power BI, Data Preprocessing

Damping Ratio Classifier: To classify the predominant Damping Ratio across different exhaust system configuration architecture based on the historical experimental damping ratios data patterns. Using the **CART model got the 96.8%** accuracy and performance on the data.

Tech Stack: Excel, Python, Machine Learning, Data Visualization, PowerBI

Education

Masters in Data Science, 2023

M.Tech in CAMDA, 2012

B.E in Mechanical, 2008

Certifications

Jain University in collaboration with Fingertips, Bangalore, KA

Manipal Institute of Technology, Manipal, KA

SDM College of Engineering and Technology, Dharwad, KA

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- **Agentic AI Fundamentals**, June 2025
 - **Introduction to Prompt Engineering**, June 2025
 - **Fundamentals of Generative AI**, December 2024
 - **Essentials of MLOps with Azure**, June 2024

Honors and Awards

Stellar Achiever Award – Leadership – Cognizant, 2024 Winner – Quality Leadership – Faurecia in 2023 Winner
– Quality Leadership – Faurecia in 2023